🌸 Can you trust a chart made from fuzzy data?

Re-creating the pictures from the research paper “Uncertainty-Aware PCA Revisited” (IEEE VIS 2025)

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The big idea, in everyday words

Flattening data is like taking a photo. A photo turns a 3D object into a flat 2D picture. A popular method called PCA does the same for data with many columns, so we can see it on a screen.

Real data is fuzzy. Measurements are never exact. A student's grade can vary from test to test, and two flowers of the same kind aren't identical.

The question: if the data is a little fuzzy, can we trust the flat picture? Or would it look totally different if the data wiggled a bit?

💡 **Think of it like a group photo where people won't stand still. Is the photo still a fair picture of the group?**

What the researchers came up with

🔮 **Picture 1: The "stability blob."** A 3D shape that shows how steady the flattening is. Thin spikes mean steady. Wide, flat parts mean shaky.

🎯 **Picture 2: The "rings."** The computer invents many slightly different versions of the data (they used one million) and takes a "photo" of each. The rings show where each item usually lands, a bit like the bullseye rings on a dartboard.

💡 **Small, separate rings mean you can trust the picture. Big, overlapping rings mean be careful.**

What I did

1. Found the researchers' code, which they shared publicly on GitHub.
2. Followed their setup instructions on my Windows laptop.
3. Ran their code on the same three datasets from the paper:

   🌺 Iris flowers (3 kinds) 🎓 Student grades (6 students) 🐸 Frog calls (4 frog families)

4. Put my pictures next to the paper's pictures to check if they match.

💡 **Re-creating someone's results is how science checks itself. If it works for me, it's more believable.**

Bumps along the way (and how I fixed them)

🔧 **Wrong version.** The code needed a newer Python than I had, so one add-on wouldn't install. I used the closest version that works with my Python.

⏳ **Too slow.** One million versions would take about 2 hours per picture. I used 100,000. The paper itself shows this looks almost the same.

🧩 **Missing piece.** The code made the 3D blob but had no way to show it, so I wrote my own viewer.

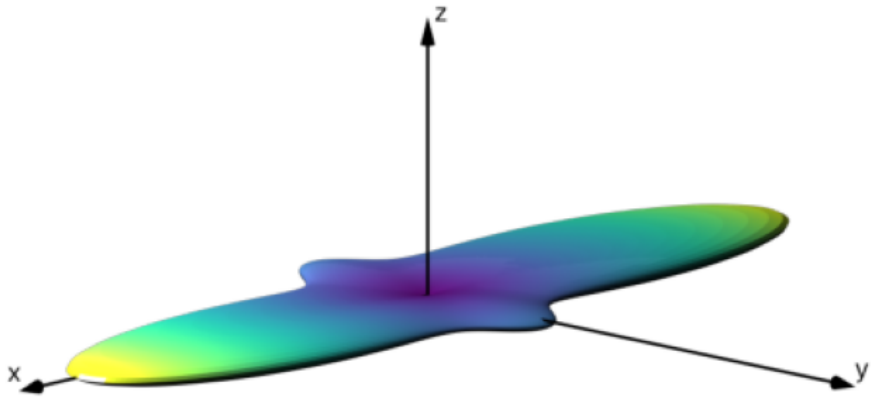
🔄 **Sideways blob.** My blobs were tipped on their side. A line in the code marked "TODO: ask if wanted" fixes it, and turning it on made the blobs match the paper.

🏷️ **Mixed-up names.** The code labels the 3 iris flower types in the wrong order. The pictures are still right; only the name tags were off.

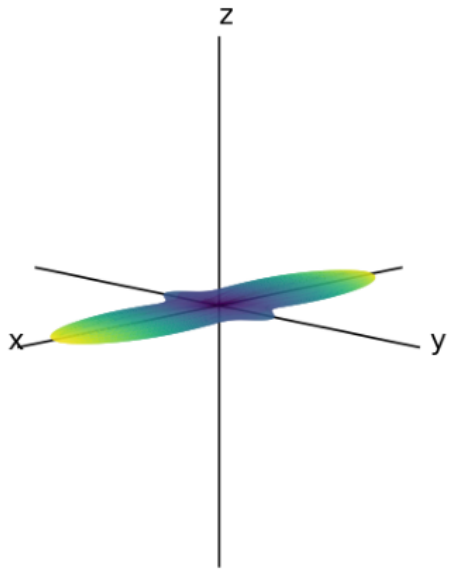
Iris flowers: the stability blob

Fig. 5 (left) – IRIS covariance stability glyph

Original (paper PDF)



Reproduced (glyph_iris_swapped)

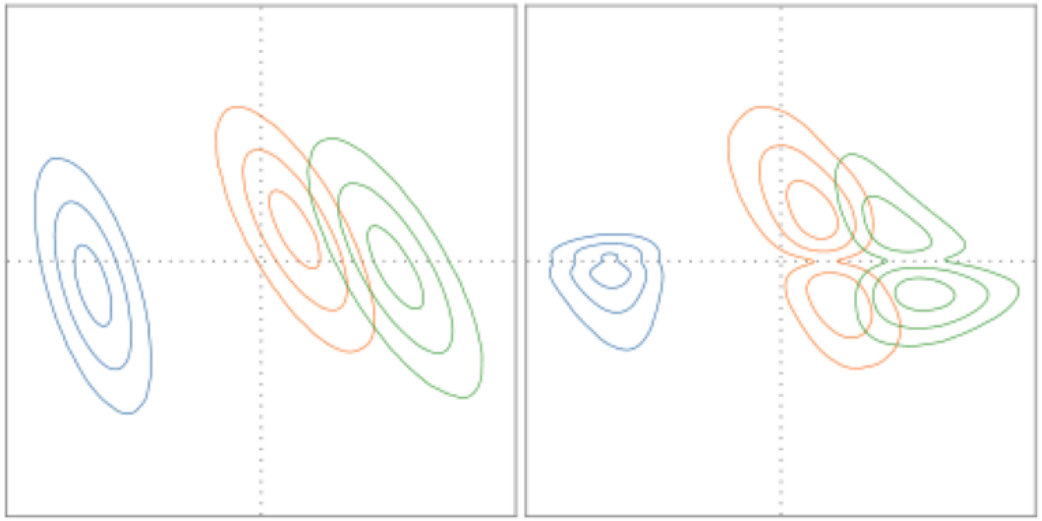


Left: the paper. Right: mine. Same long, flat shape, so the flower picture is less steady than it looks.

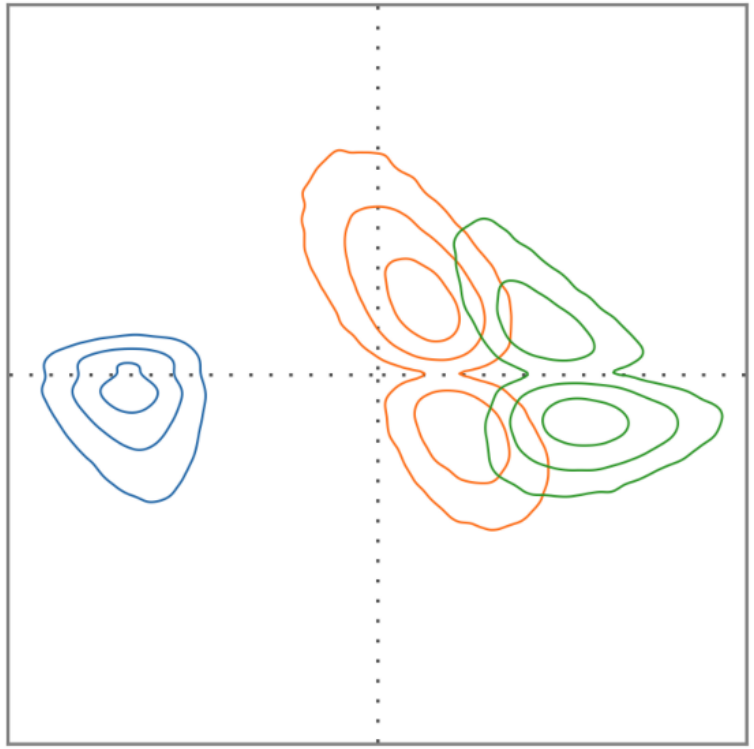
🌺 Iris flowers: where each kind lands

Fig. 5 (right) – IRIS sampling-based projection

Original (paper PDF)



Reproduced (projection_iris_U100000)

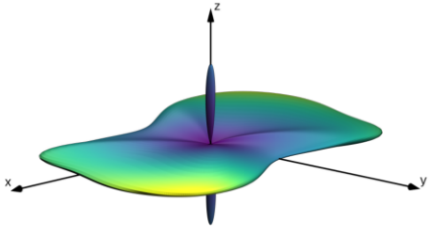


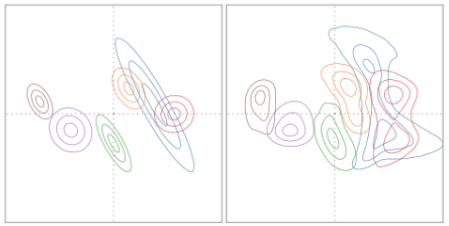
💡 **Blue (Setosa) sits alone, so it’s easy to tell apart. Orange and green overlap, so those two kinds get mixed up.**

🎓 Student grades

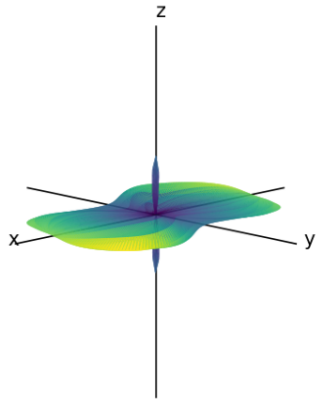
Fig. 8 – STUDENT glyph (left) and projection (right)

Original (paper PDF)

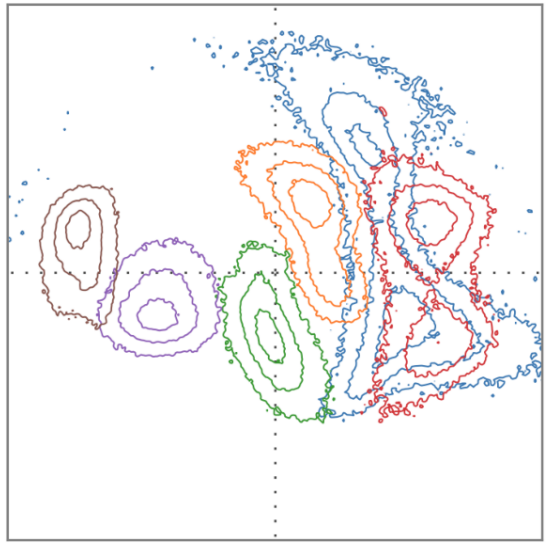




Reproduced (glyph_students_swapped)



Reproduced (projection_students_U100000)

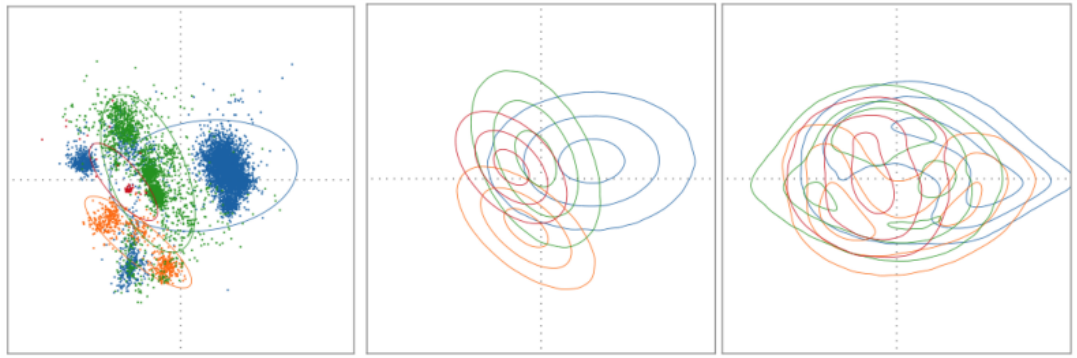


💡 **Thin spike plus a wide flat disc: the main pattern is steady, but the second one isn’t. So the students’ positions are shakier than older methods showed.**

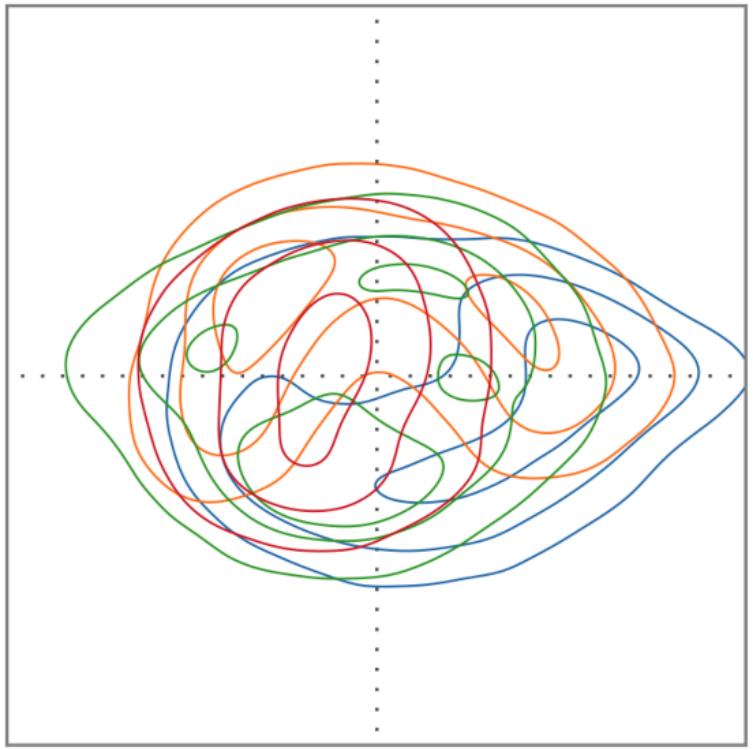
🐸 Frog calls

Fig. 9 (right) – ANURAN CALLS sampling-based projection

Original (paper PDF)



Reproduced (projection_anuran_U100000)



💡 **All the rings pile on top of each other. A normal flat chart of this data would be misleading!**

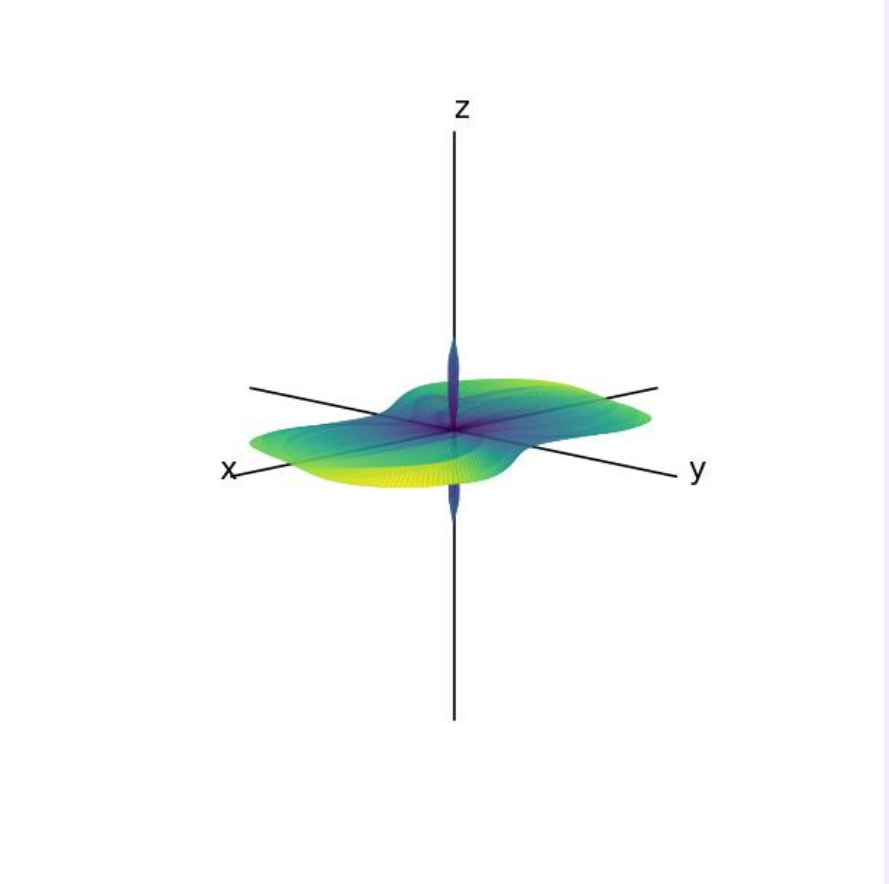
✨ Bonus: I made it move

The paper only has still pictures, so I built a website where you can:

🔄 Spin the 3D blob with your mouse

🎬 Watch the “photo” change, one version of the data at a time. Steady items barely move, and shaky ones jump around.

🔗 ponzek.github.io/CSP5745WK9AS2



What I learned

✅ It worked. All 4 pictures I tried match the paper.

📏 The small differences came from using fewer versions (100,000 instead of 1,000,000) and a different 3D viewer.

🧠 Big lesson: a chart can look sure of itself even when the data behind it is fuzzy. Showing the fuzziness helps people avoid wrong conclusions.

🤝 The researchers shared their code and data, which made this possible. That's what good science looks like.

💡 **Thank you!** 💜 **Questions?**